LESSON 10.3

Math is Playful

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use patterns and relationships to make sense of the Tower of Hanoi puzzle and think logically about its solutions.

WHY THIS MATTERS The Tower of Hanoi is a puzzle invented in 1883. It has three rods and a stack of discs, smallest on top to largest on the bottom. The number of discs can vary, but the number of rods does not — and hiding inside it is one of the prettiest patterns in mathematics.

REMEMBER

Math is Playful

Algorithmic Thinking & Recursive Patterns The objective is to move the whole stack of discs to another rod. The rules: move only one disc at a time, move only the top disc from a stack, and never place a larger disc on top of a smaller disc.

1. Deconstruct complex multi-step problems into smaller, repeatable sub-tasks.
2. Trace patterns from simple cases to formulate general mathematical rules.
3. Verify algorithmic rules across edge cases and increasing problem sizes

Words: puzzle *(rompecabezas)* · pattern *(patrón)* · pattern rule *(regla del patrón)* · predict *(predecir)* · organize *(organizar)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each move: legal or against the rules?

Groups: Legal move · Against the rules

_____ Move the top disc of any stack

_____ Place a small disc on a bigger disc

_____ Move exactly one disc at a time

_____ Move two discs at once

_____ Place a big disc on a smaller disc

_____ Pull a disc out from the middle of a stack

HINT Only the TOP disc of a stack can move, and only one at a time.

2 **Complete the table. Show how you found each value.**

Use the doubling pattern to complete the table of fewest moves.

Discs	Pattern: double the previous, add 1	Fewest moves
3		
4		
6		

HINT Each new disc means: solve the smaller tower, move the big disc, solve the smaller tower again.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Will the fewest number of steps for a Tower of Hanoi puzzle always be an ODD number?

(A) Yes — doubling any whole number gives an even number, and adding 1 makes it odd

(B) No — every other tower size gives an even number of steps

(C) Yes — but only because 7 happens to be odd

(D) No — the steps alternate between odd and even

HINT Look at the whole list: 1, 3, 7, 15, 31.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

How many lines of symmetry does a square have?

(A) 1

(B) 2

(C) 4

(D) 8

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A shape has bilateral symmetry. This means…

(A) it has four equal sides

(B) one line splits it into two mirror halves

(C) it can be turned in a circle

(D) it has no straight edges

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Which capital letter has a vertical line of symmetry?

(A) R

(B) A

(C) P

(D) F

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a move you called legal and your partner called a rule-break. What is the rule in your own words, and does it still hold on a move you both agree about?

SENTENCE STARTER The rule is ___, so ___ is legal / breaks it because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 10.3

Math is Playful

Set B · Version B · Second form · core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use patterns and relationships to make sense of the Tower of Hanoi puzzle and think logically about its solutions.

WHY THIS MATTERS The Tower of Hanoi is a puzzle invented in 1883. It has three rods and a stack of discs, smallest on top to largest on the bottom. The number of discs can vary, but the number of rods does not — and hiding inside it is one of the prettiest patterns in mathematics.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which statement best shows what this lesson means by 'math is playful'?

(A) Math is playful only when there are no rules to follow

(B) Puzzles like the Tower of Hanoi can only be solved by trial and error

(C) We play with math by imagining possible solutions and looking for patterns that help us think logically about puzzles

(D) Playing with math means guessing until something works

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

How many lines of symmetry does a regular hexagon have?

(A) 2

(B) 3

(C) 6

(D) 12

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

You solved the small towers by hand: 1 disc takes 1 step, 2 discs take 3 steps, 3 discs take 7 steps. Now someone hands you a five-disc tower and asks for the fewest steps — but you do not have time to play it out. The table shows: 1 disc = 1 step, 2 discs = 3 steps, 3 discs = 7 steps. What is the pattern rule from one row to the next?

(A) Add 2 to the previous number of steps

(B) Multiply the previous number of steps by 3

(C) Double the previous number of steps and add 1

(D) Add the disc number to the previous steps

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

You solved the small towers by hand: 1 disc takes 1 step, 2 discs take 3 steps, 3 discs take 7 steps. Now someone hands you a five-disc tower and asks for the fewest steps — but you do not have time to play it out. Using the rule "double the steps and add 1," how many steps does a 4-disc tower take, starting from the 3-disc value of 7 steps?

(A) 8

(B) 14

(C) 15

(D) 21

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

You solved the small towers by hand: 1 disc takes 1 step, 2 discs take 3 steps, 3 discs take 7 steps. Now someone hands you a five-disc tower and asks for the fewest steps — but you do not have time to play it out. Continuing the pattern from 15 steps for a 4-disc tower, how many steps does a 5-disc tower take?

(A) 16

(B) 30

(C) 31

(D) 45

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a move you called legal and your partner called a rule-break. What is the rule in your own words, and does it still hold on a move you both agree about?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help